Darsh Suhas Kanikar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.18
Alpha Junior College of Science and Commerce	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education (MSBSHSE))	2021	89.83%
Ryan International School	Class X (Indian Certificate of Secondary Education)	2019	92.28%

PROJECTS

Secure Money Transfer Web Application (TransactiWar) | Feb 2026 - Apr 2026

Mentor: Dr. Saurabh Kumar, CSE - IIT Hyderabad | GitHub

- Built a framework-less secure money-transfer web app (PHP, PostgreSQL, Nginx, Docker) implementing Argon2id hashing, CSRF tokens, prepared statements, output escaping, rate limiting, and hardened sessions to mitigate OWASP Top 10 risks.
- Ensured transfer integrity with atomic transactions and row-level locking; added audit logging, secure image uploads, and HTTPS-only deployment (TLS 1.2/1.3, HSTS, CSP) in non-root, read-only containers.

Anti-Emulation Detection Based on Sensor Data | Sep 2025 - Nov 2025

Mentor: Dr. Saurabh Kumar, CSE - IIT Hyderabad | GitHub

- Collected 19-channel smartphone sensor data (accelerometer, gyroscope, magnetometer, orientation, gravity) at 100 Hz across 4 activities and built a preprocessing pipeline (alignment, interpolation, resampling, MinMax scaling).
- Benchmarked TimeGAN, RCGAN, VAR, SMOTE, augmentation, and PSD synthesis; VAR best preserved temporal structure and cross-sensor correlations, validated via cadence, RMS acceleration, correlation-matrix differences, and PSD analysis.
- In a red-team exercise, the opposing team's detector failed to flag our VAR-generated logs as synthetic .

Advancements in Latent Diffusion Models for Multi-Scale Image Generation | Aug 2025 - Nov 2025

Mentor: Dr. C. Krishna Mohan, CSE - IIT Hyderabad | GitHub

- Reproduced ZoomLDM (CVPR 2025), a multi-scale latent diffusion model for histopathology (UNI encoder, cross-magnification summarizer, cross-attention U-Net, DDIM, classifier-free guidance), on a 20-image subset under limited compute.
- Proposed BioFidelity, a histopathology-aware metric combining SSIM, nuclei-count consistency, GLCM texture correlation, and UNI-feature FID, addressing FID's reliance on ImageNet features.

AI Resume Analyzer | Jun 2024 - Apr 2025

Mentor: Dr. Lalit Purohit, IT - SGSITS Indore | Review paper

- Built a RAG-based resume screening system (LangChain, FAISS, GPT-4o-mini, Streamlit) that decomposes job descriptions into sub-queries, retrieves and reranks resume chunks, and returns a ranked shortlist with justifications.
- Published a review paper on AI-based resume analyzers at the International Conference on Computer Science and Communication Engineering (ICCSCE-2025).

Fruit Quality Assessment using Supervised Learning | Jun - Jul 2024

Mentor: Dr. Gaurav Mishra, CSE - SVNIT Nagpur | Research Paper

- Built an image-classification pipeline on the FruitsGB dataset (12,000 images, 12 classes) using color-histogram, HOG, and mean-color features (35,111 dimensions) with an 80/20 split.
- Benchmarked 9 ML/DL models plus YOLOv8; Random Forest performed best at 97.75% test accuracy (98% F1), ahead of SVM (96.79%) and Logistic Regression (96.58%).
- Published a research paper at the 2nd IEEE International Conference on Cognitive, Green and Ubiquitous Computing (IC-CGU-2025)**.**

SCHOLASTIC ACHIEVEMENTS

- Certificate in Programming, Data Structures and Algorithms using Python Jan-March 2023 issued by NPTEL, secured 80% marks, was among top 5% of candidates.

- Successfully completed TCS iON Career Edge - Young Professional course. December 2022

- Codechef rank: 1387

- LeetCode- Solved 147 problems (82 Easy, 57 Medium, 8 Hard)

RELEVANT COURSES

Natural Language Processing, Representation Learning, Foundations of Machine Learning, Data Mining, Practical Malware Analysis ,Network Security

POSITIONS OF RESPONSIBILITY

Teaching Assistant in Data Structures and Applications | July to Nov 2026

Teaching Assistant in Operating Systems - II | Jan to Apr 2026

Actively volunteered in placement activities - 2025

Teaching Assistant in Data Structures and Applications (ID 2230) | July to Nov 2025

SKILLS

Operating Systems : Windows ,Linux

Programming Skills : Python, C,C++, Java

AI/ML: Natural Language Processing (NLP), Traditional ML models, Data Preprocessing